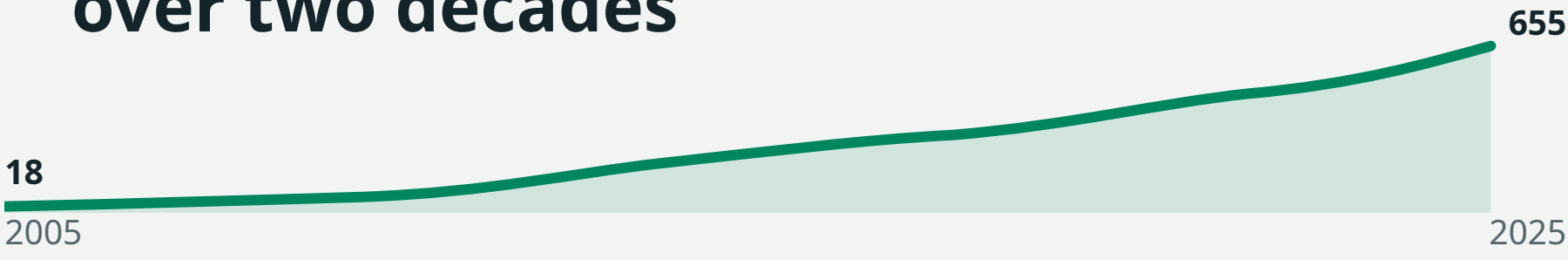


Perturbation evidence is multiplying faster than it can be organised

A single pooled genome-scale Perturb-Seq dataset can **simultaneously profile tens of millions of cells across hundreds of thousands of perturbations**. That scale could reveal robust biology across studies, but identifiers, metadata, assays and processing pipelines differ. The Perturbation Catalogue aims to unify datasets across modalities and provide a shared access path for this evidence.

36× growth in perturbation papers over two decades



Europe PMC title/abstract query; 24 Aug 2026.
Scale example: FirstWord Pharma, story 7139212.

Our approach turns scattered datasets into a unified resource

1,237
datasets

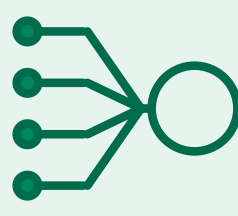
19,235
targets

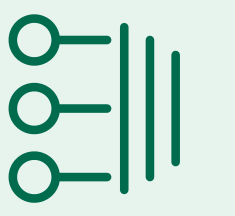
36
tissues

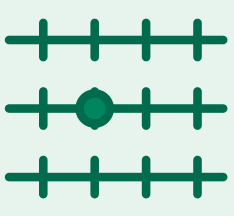
34
cell types

1,200
cell lines

169
diseases

**CRISPR screens · 1,201 datasets**
Perturbed gene → phenotype / fitness changes.

**Perturb-Seq · 26 datasets**
Perturbed gene → scRNA-seq, cell-wide transcriptome changes.

**MAVE · 10 datasets**
Combinatorial variant library → sequence-function maps.

We extensively curate metadata...

84
meta-
data
fields

Study & provenance
Title, authors, year, URI, licence

Model system
Organism, tissue, cell line, disease

Ensembl gene IDs

Perturbation & library
Targets, guides, library, variants

Readout & processing
Assay, technology, kit, reference

Ontology term mapping

And reprocess data from scratch

SRA / FASTQ
Raw reads + study-specific probes

→ **Cell × gene counts**
Shared reference + gene annotation

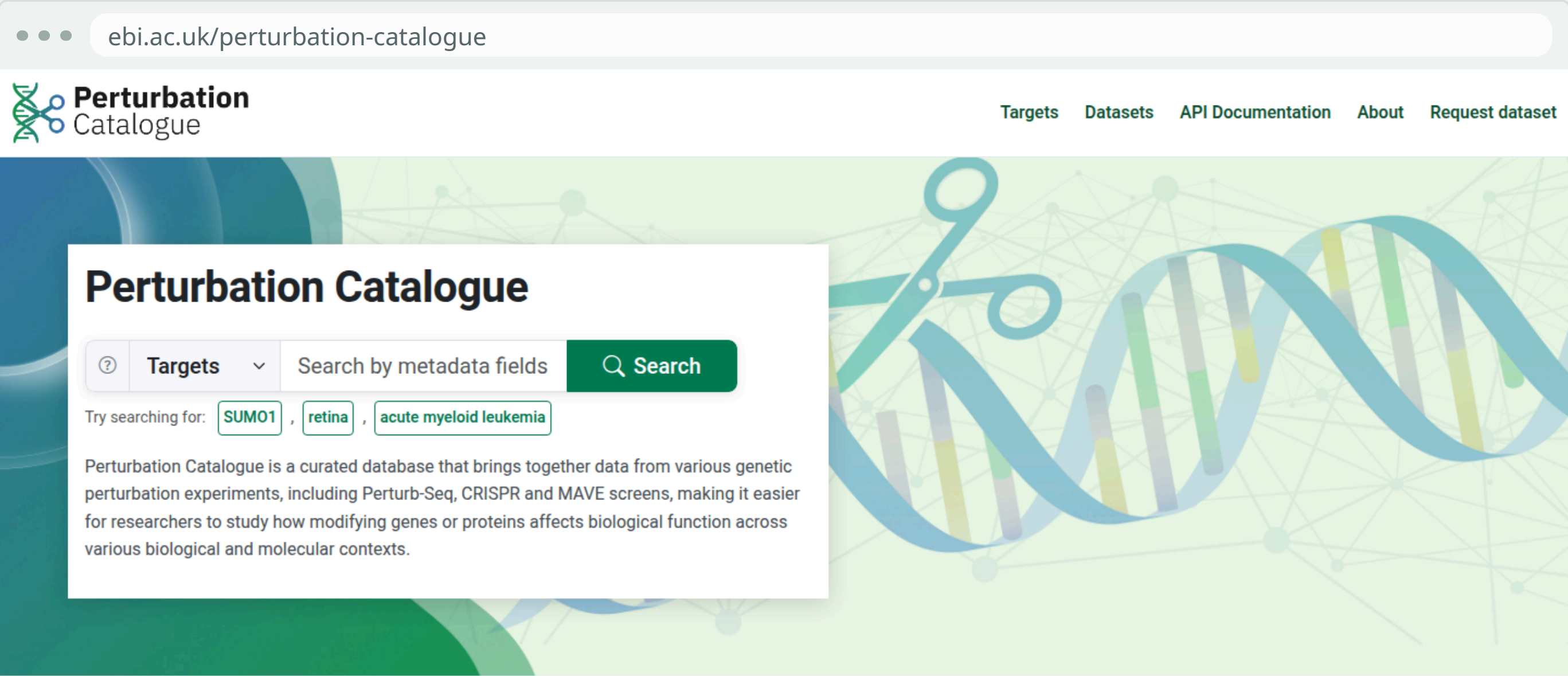
→ **Guide assignment**
Map cells to perturbations

→ **QC + filtering**
Final analysis-ready H5AD

Five Perturb-Seq datasets are live. Perturbation-control comparisons feed DEA and Hallmark GSEA. More datasets and modalities coming soon.

Browse the Catalogue

Search → Context → Filter → Explore



Perturbation Catalogue at a glance
Explore high-level statistics across the catalogue before diving into specific targets.

DATASETS
1237 (spanning 2014 - 2025)

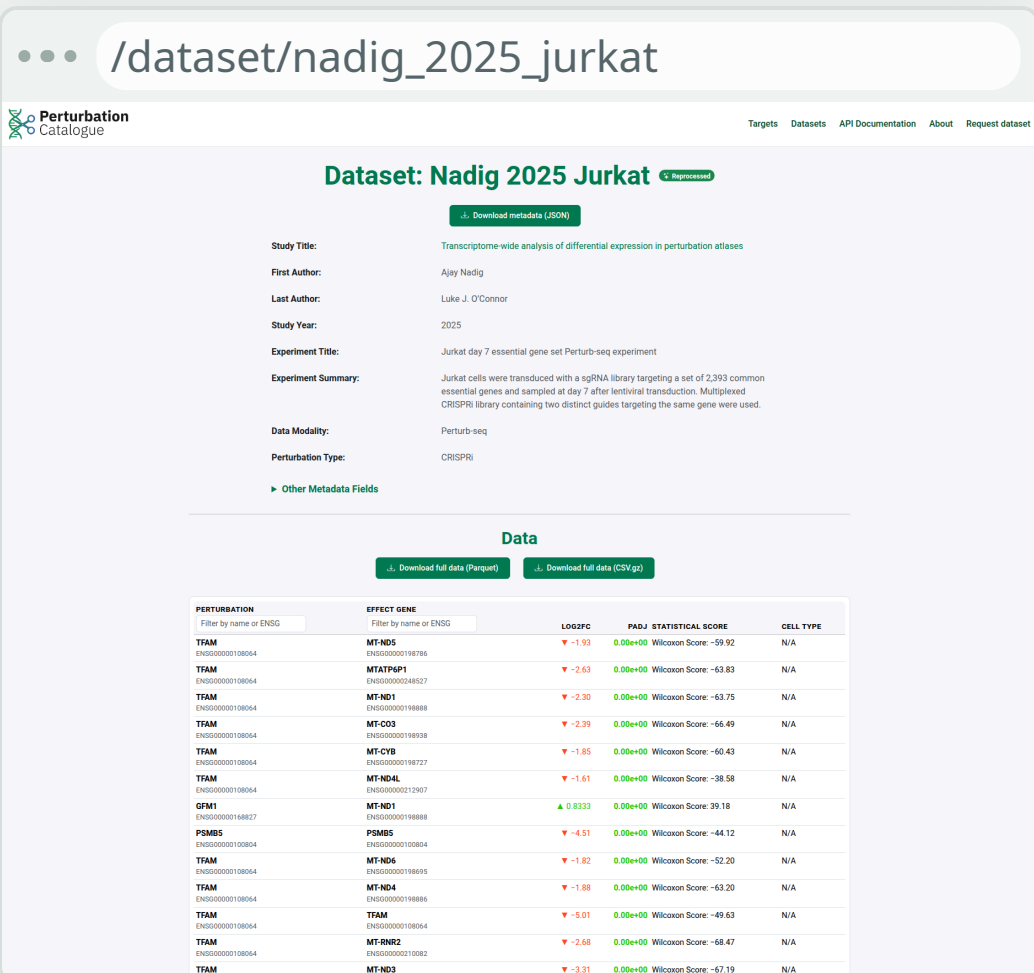
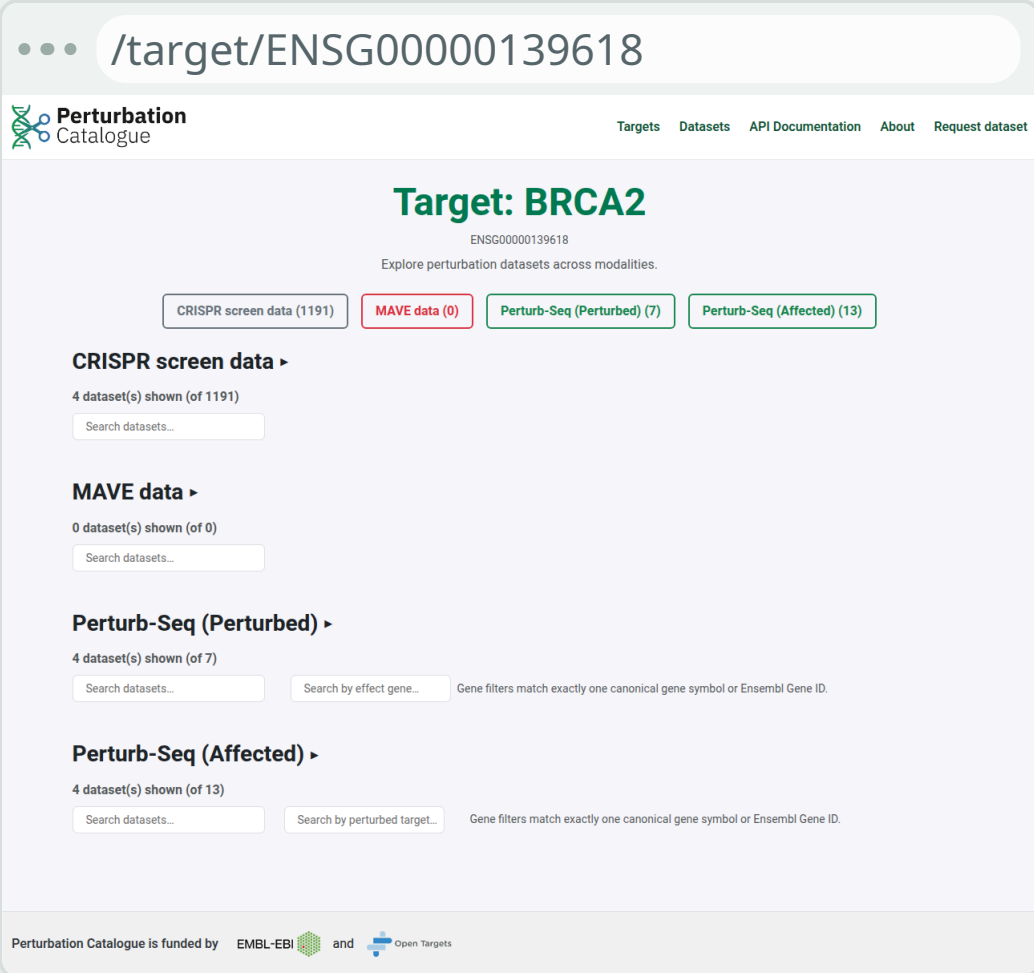
TARGETS
19235

UNIQUE TISSUES
36

UNIQUE CELL TYPES
34

UNIQUE CELL LINES
1200

UNIQUE DISEASES
169



Access records via API

Structured JSON for custom pipelines, reproducible analysis and machine learning.

```
GET /v1/perturb-seq/nadig_2025_jurkat/search?perturbation_gene_name=TFAM&limit=1
```

```
{
  "perturbation": {
    "perturbed_target_ensg": "ENSG00000108064",
    "gene_symbol": "TFAM"
  },
  "effect": {
    "effect_gene_ensg": "ENSG00000210082",
    "gene_symbol": "MT-RNR2",
    "log2fc": -2.682
  }
}
```

Download entire datasets

Data is available in **Parquet** and **CSV.gz**, and metadata as **JSON**.

Stay tuned for more exciting things coming soon

**Extract structured metadata faster**
AI-assisted drafting with curator review and validated ontology mapping.

**Reanalyse more designs**
More kits, experimental structures and independent quality metrics.

**Publish regular, versioned releases**
Explicit provenance and stable snapshots for reproducible downstream use.

**Reanalyse MAVE data**
Enrich variant-to-function coverage with harmonised scores.

Visit: ebi.ac.uk/perturbation-catalogue Contact: perturbation-catalogue-help@ebi.ac.uk

